

VENOMSENSE

LOCALIZED BITE-SITE TOXIN DETECTION SYSTEM

EXECUTIVE SUMMARY

VenomSense is an IoT-enabled, portable medical wearable designed to instantly detect snake venom directly from the bite-site wound. It empowers rural healthcare workers to identify venomous bites within minutes, preventing deaths caused by delayed treatment and "Dry Bite" panic.

1. THE PROBLEM STATEMENT

50,000+

ANNUAL SNAKEBITE MORTALITIES IN INDIA

>50%

RECORDED AS NON-VENOMOUS DRY BITES

India accounts for over 50,000 snakebite mortalities annually. The crisis is compounded by two major factors: delay in medical intervention and the "Dry Bite" phenomenon. Over 50% of bites are non-venomous, yet victims succumb to fear-induced cardiac arrests. Furthermore, doctors hesitate to administer Anti-Snake Venom (ASV) due to the risk of severe anaphylactic shock if venom is not actually present in the patient's body.

2. LOCALIZED BITE-SITE TESTING

Instead of drawing systemic blood (which takes time for venom to reach), VenomSense samples the lymphatic tissue fluid directly from the fang puncture marks. The system completely eliminates the guesswork and wait-times through a cheap, disposable Cartridge and a reusable Electronic Smart Reader.

3. OPERATIONAL WORKFLOW

01

THE CARTRIDGE

A disposable bell-shaped cartridge is placed over the snake's fang marks. A button press creates micro-suction, safely extracting wound exudate (venom residue) without any painful needles.

02

THE BIOCHEMICAL TRAP

The fluid enters a Lateral Flow Assay (LFA) strip inside the cartridge. Gold-nanoparticle tagged antibodies act as a "Lock" that exclusively binds to the "Key" (Indian snake venom proteins).

03

THE SMART MACHINE

The cartridge is inserted into the VenomSense device (powered by ESP32). A TCS34725 Optical Color Sensor scans the strip for microscopic colorimetric changes invisible to the naked eye.

04

DIGITAL RESULT

The ESP32 processes the optical data and outputs a verified clinical result on the OLED screen within 60 seconds (e.g., "Positive: Toxin Detected" or "Negative: Dry Bite").

4. TECHNICAL SPECIFICATIONS

HARDWARE CORE: ESP32-C3 Microcontroller with onboard Wi-Fi/BLE for IoT capabilities.

SENSOR MODULE: TCS34725 RGB Color Reflectance Photometer.

DISPLAY: 0.96 inch I2C OLED display for clear, rural-friendly reading.

BIOTECH ASSAY: Colloidal Gold Lateral Flow Assay (LFA) for Big Four Elapids & Vipers.

5. SOCIAL & ECONOMIC IMPACT

AFFORDABILITY: The reusable hardware costs under ₹1,500, and mass-produced LFA cartridges will cost ~₹50, replacing expensive lab diagnostic tests.

ZERO FALSE POSITIVES: The highly specific antibody mechanism ensures the device only triggers for actual venom proteins, not sweat, normal blood, or acidic fluids.

RESOURCE OPTIMIZATION: Ensures expensive ASV vials are utilized only when absolutely necessary, saving hospital resources and preventing allergic shock fatalities.